

Online Retail II Dataset – Exploratory Data Analysis (EDA)

AI/ML Internship – Week 1 Report

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1. Introduction

The purpose of this project is to perform Exploratory Data Analysis (EDA) on the Online Retail II Dataset. The analysis helps understand customer purchasing behavior, identify data quality issues, discover the best-selling products, analyze country-wise sales performance, and examine monthly revenue trends. These insights support better business decision-making and provide a foundation for developing an effective recommendation system.

2. Dataset Description

The Online Retail II Dataset contains real-world transaction records from a UK-based online retailer between 2009 and 2011.

The dataset includes the following information:

- Invoice Number
- Stock Code
- Product Description
- Quantity
- Invoice Date
- Unit Price
- Customer ID
- Country

Screenshot:

```
#START THE TASK GIVEN BY IT SIMPLERA INSTITUTE

#IMPORT THE IMPORTANT LIBRARY FOR EXPLORATORY DATA ANALYSIS (EDA)

import numpy as np
import pandas as pd
import openpyxl as op
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px

#LOAD THE DATA

df=pd.read_excel('online_retail_II.xlsx')

#DISPLAY THE FIRTS FEW ROW TO USE (head) METHOD

df.head()
```

	Invoice	StockCode	Description	Quantity	InvoiceDate	Price	Customer ID	Country
0	489434	85048	15CM CHRISTMAS GLASS BALL 20 LIGHTS	12	2009-12-01 07:45:00	6.95	13085.0	United Kingdom
1	489434	79323P	PINK CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
2	489434	79323W	WHITE CHERRY LIGHTS	12	2009-12-01 07:45:00	6.75	13085.0	United Kingdom
3	489434	22041	RECORD FRAME 7" SINGLE SIZE	48	2009-12-01 07:45:00	2.10	13085.0	United Kingdom
4	489434	21232	STRAWBERRY CERAMIC TRINKET BOX	24	2009-12-01 07:45:00	1.25	13085.0	United Kingdom

3. Dataset Overview

After loading the dataset, I examined its structure, dimensions, column names, and data types.

Results

- Dataset Shape: $((525461, 8))$
- Number of Rows: (525461)

- Number of Columns: (8)

Screenshot of shape columns and datatypes

```
df.shape
(525461, 8)

2 column names and data types

df.columns
Index(['Invoice', 'StockCode', 'Description', 'Quantity', 'InvoiceDate',
      'Price', 'Customer ID', 'Country'],
      dtype='object')

df.dtypes
Invoice      object
StockCode    object
Description   object
Quantity      int64
InvoiceDate  datetime64[ns]
Price        float64
Customer ID  float64
Country      object
dtype: object
```

4. Data Quality Analysis

Missing Values

The dataset was checked to identify missing values in each column.

Invoice	0
StockCode	0
Description	2928
Quantity	0
InvoiceDate	0
Price	0
Customer ID	107927
Country	0

Screenshot

```
#FIND MISSING VALUE AND DUPLICATE

df.isnull().sum()

[11]

... Invoice      0
    StockCode    0
    Description  2928
    Quantity     0
    InvoiceDate   0
    Price        0
    Customer ID  107927
    Country      0
    dtype: int64
```

Observation

Missing values were found in some columns. These values will need to be handled during the data preprocessing stage before applying machine learning algorithms.

Duplicate Rows

The dataset was checked for duplicate records.

Screenshot

```
CHECK THE DUPLICATE VALUE IN DATASET

df.duplicated().sum()

[12]

... np.int64(6865)
```

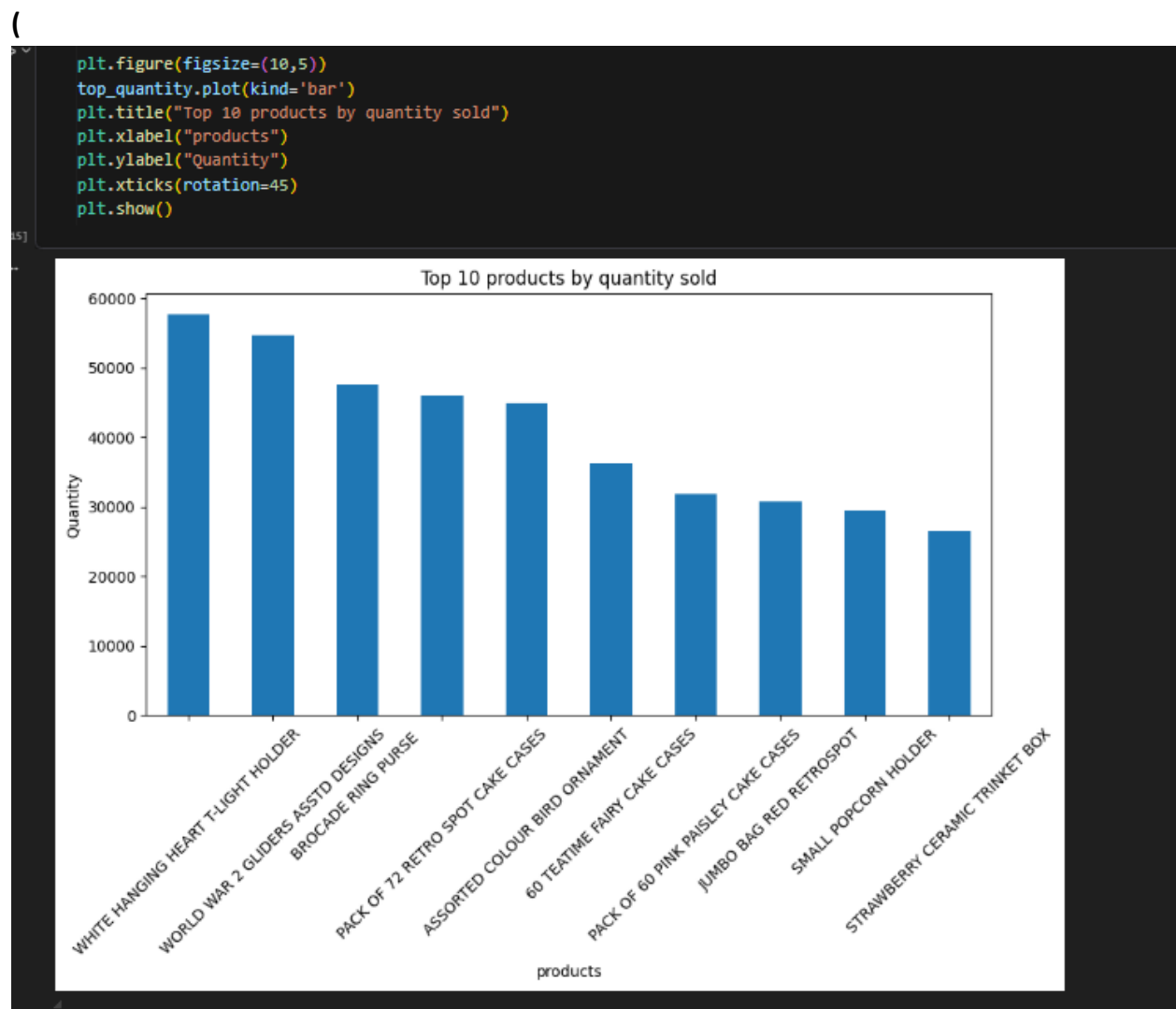
Observation: Duplicate records were identified. These records can affect the analysis if they are not removed during data preprocessing.

5. Exploratory Data Analysis

Top 10 Best-Selling Products by Quantity

A bar chart was created to identify the products sold in the highest quantities.

Screenshot



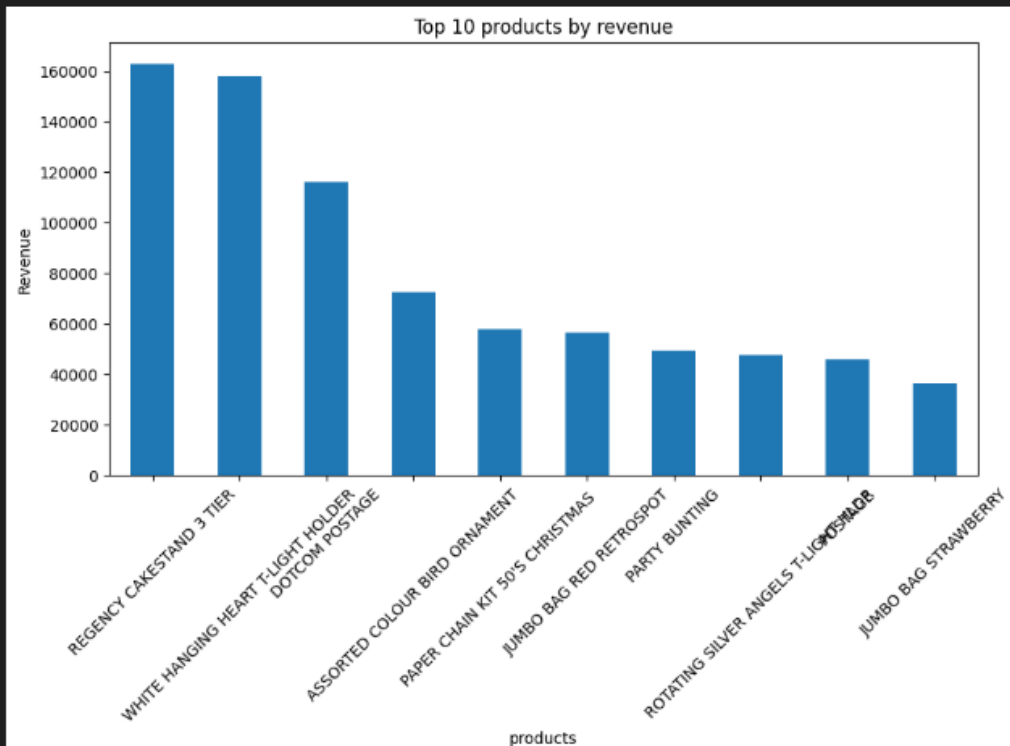
Explanation: The chart shows the products that customers purchased most frequently.

Top 10 Products by Revenue

A revenue analysis was performed to identify the products generating the highest income.

Screenshot

```
plt.figure(figsize=(10,5))
top_revenue.plot(kind='bar')
plt.title("Top 10 products by revenue")
plt.xlabel("products")
plt.ylabel("Revenue")
plt.xticks(rotation=45)
plt.show()
```



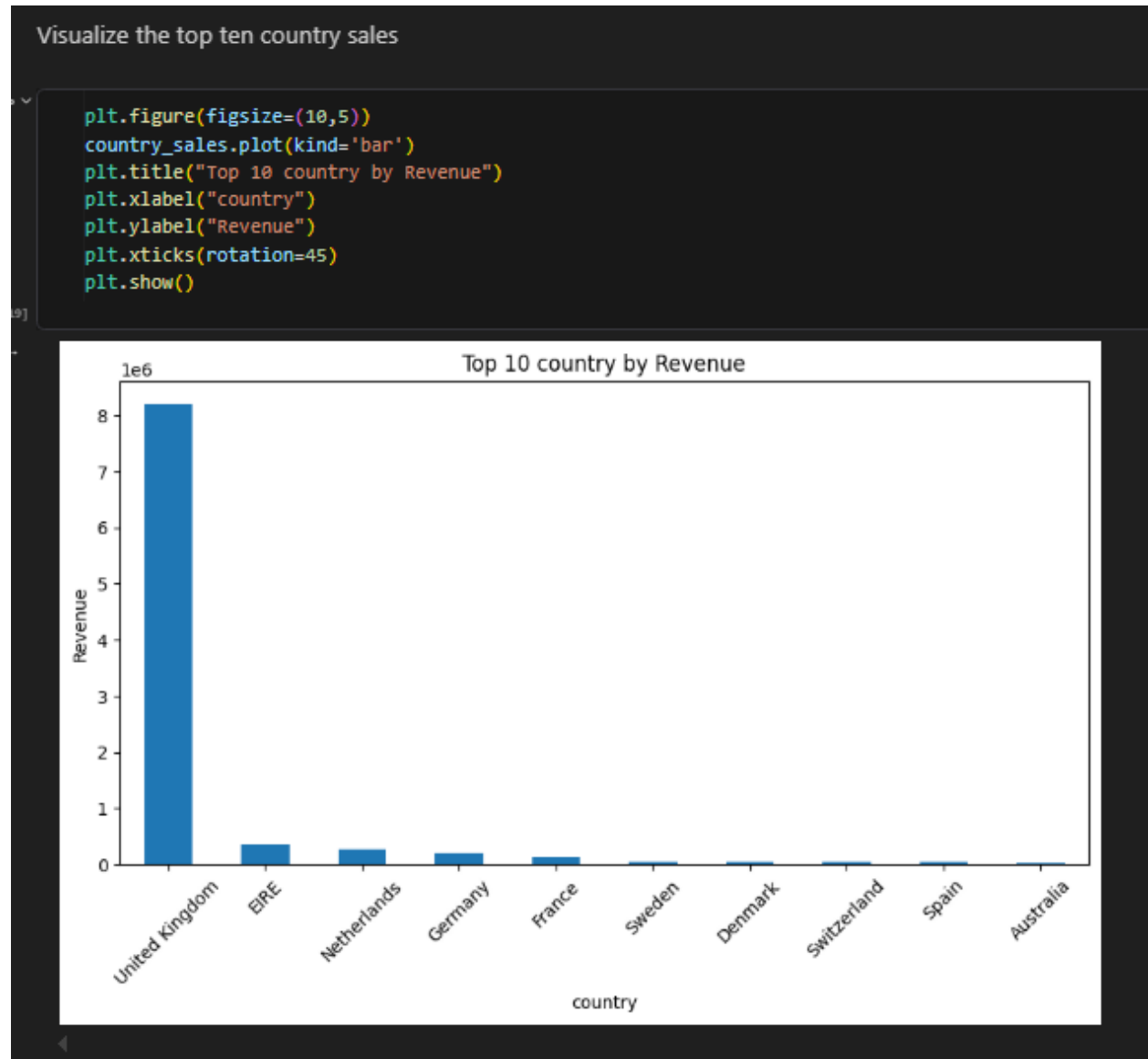
Explanation

The graph highlights the products that generated the highest revenue for the company.

Sales Performance by Country

Country-wise sales analysis was performed.

Screenshot



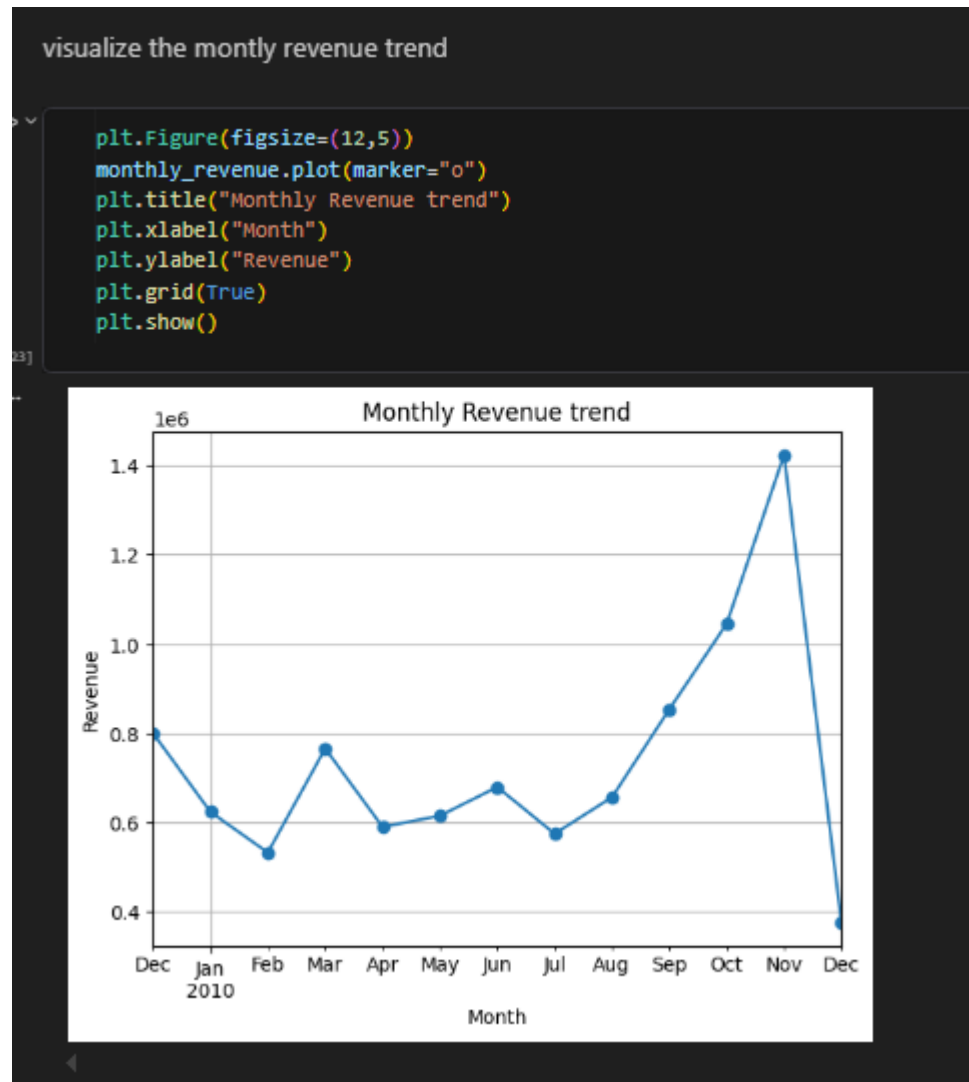
Explanation

This chart compares the sales performance of different countries and identifies the countries contributing the highest revenue.

Monthly Revenue Trend

Monthly revenue was calculated and visualized using a line chart.

Screenshot



Explanation

The graph shows changes in revenue over time and helps identify seasonal sales patterns.

Correlation Heatmap

A correlation heatmap was created to study the relationship between numerical variables.

Screenshot



Explanation

The heatmap illustrates the strength of relationships among the numerical features in the dataset.

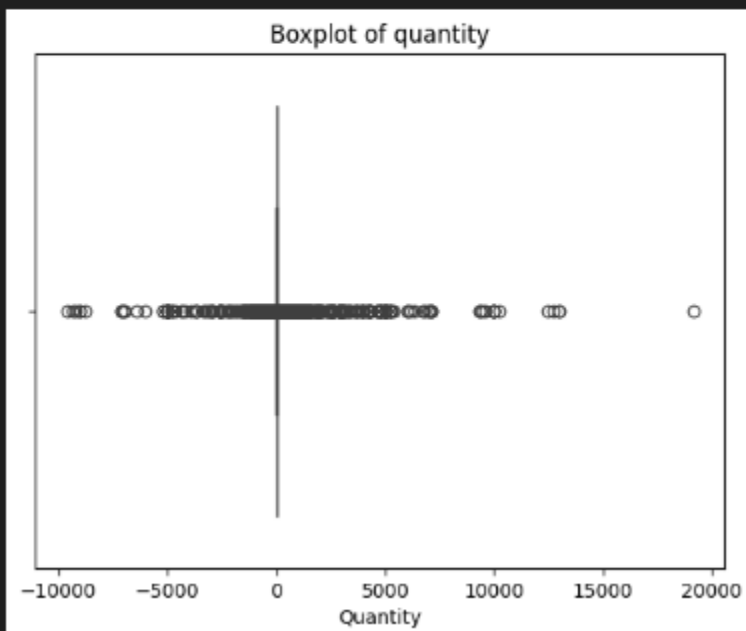
Outlier Detection

Box plots were used to detect outliers in the numerical columns.

Screenshot

```
sns.boxplot(x=df["Quantity"])  
plt.title("Boxplot of quantity")
```

Text(0.5, 1.0, 'Boxplot of quantity')



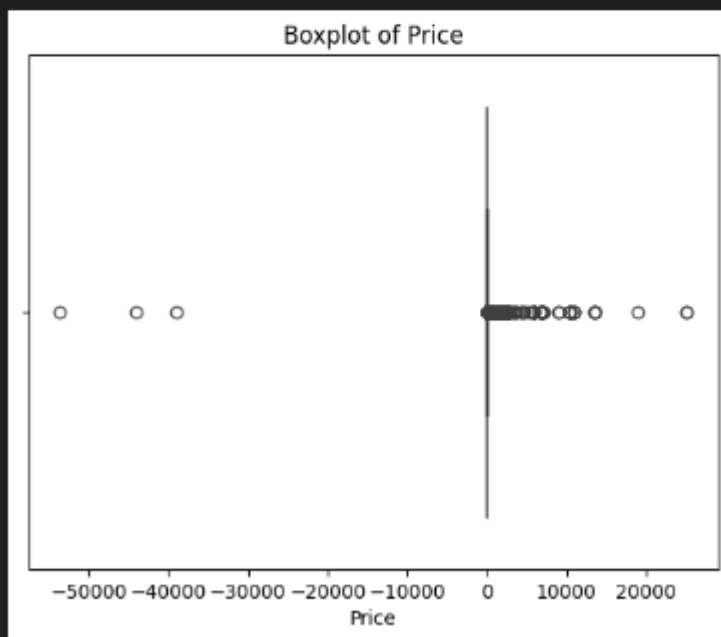
Boxplot of price

Screenshot

boxplot for price

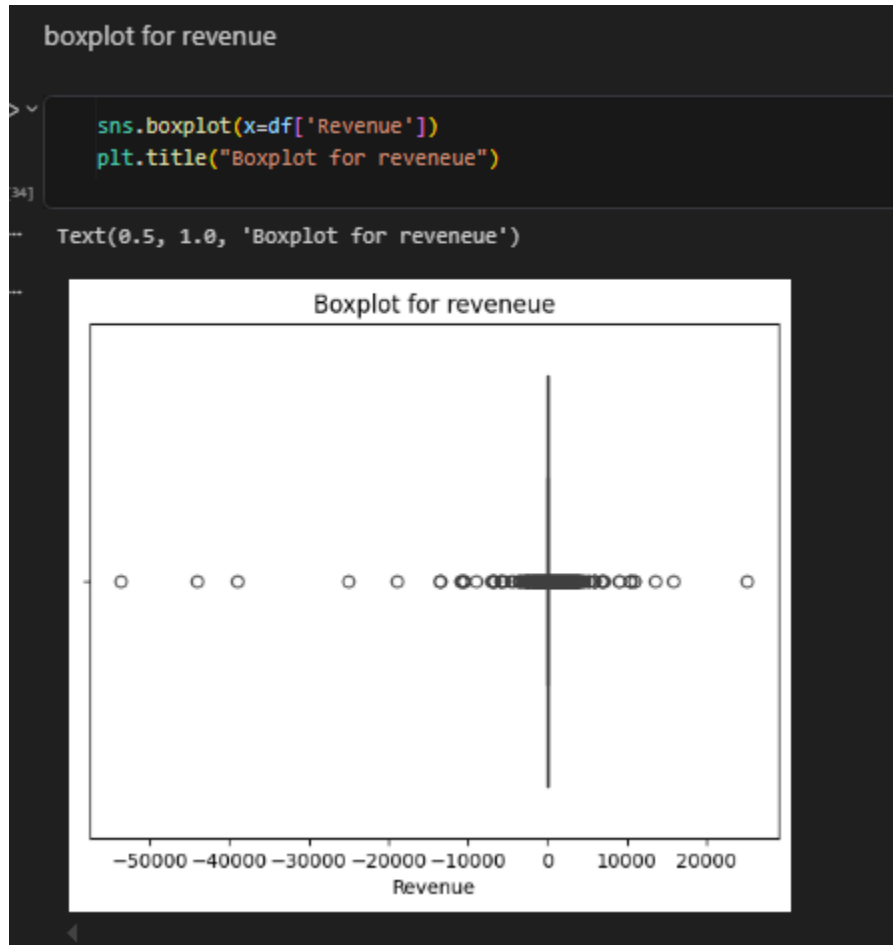
```
sns.boxplot(x=df['Price'])  
plt.title("Boxplot of Price")
```

Text(0.5, 1.0, 'Boxplot of Price')



Box plot of Revenue

Screenshot



Explanation

The box plots reveal the presence of extreme values that may influence future analysis and machine learning models.

6. Business Insights

1. The highest-selling products contributed significantly to the total quantity sold.
 2. A small number of products generated a large portion of the total revenue.
 3. Some countries contributed much more revenue than others.
 4. Monthly sales showed fluctuations, indicating seasonal purchasing behavior.
 5. The dataset contains missing values, duplicate records, and outliers that should be handled before building predictive models.
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7. Conclusion

This Exploratory Data Analysis successfully examined the Online Retail II Dataset and provided valuable insights into customer purchasing behavior, product performance, and sales trends. The analysis also identified important data quality issues such as missing values, duplicate records, and outliers. These findings will help prepare the dataset for future machine learning tasks and support the development of an effective recommendation system.

8. Tools and Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Plotly
 - Jupyter Notebook
-

9. References

- Online Retail II Dataset (UCI Machine Learning Repository)
 - Python Documentation
 - Pandas Documentation
 - Matplotlib Documentation
 - Seaborn Documentation
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10. GitHub Repository

GitHub Repository Link:

<https://github.com/farhan-magsi/week-1-task-from-ITSimplera-institute.git>